

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. MTB-509: Relativity

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1. (a) Show that Newton's second law of

(a) is invariant under special ; Galilean transformations. 4 :

(b) Show that Lorentz transformation equation are superior to Galilean transformation. °4

(c) 'At what speed should a clock be . moved so that it may appear to lose , 1 minute in each hour? [2]

; ; | (a) The length of a rocket ship is 100m = |; on the ground. When it is in flight)

(c) length observed on the ground is) 99 m. Calculate its speeds. = 2 |

(d) Show that if 12 is the rest volume of :

(e) cube, then i3 jl = v^2/c^2 is the); volume viewed from a reference | frame moving in a direction parallel 'to an edge

(e) Show that the circle $x^2 + y^2 = a^2$ in the frame S appear to be an ellipse |

(e) S' which is moving with velocity v . relative to S along x-axis. [3] . (g) Explain Minkowski — four-dimensional space-time. [3] (h) Define the 4-D flat space 'time' metric. What is the use of this : metric for 4 - D vectors? . [2]

Question 2. (a) A Cartesian frame OX'YZ rotates . about the axis OZ with uniform angular velocity w relative to the inertial frame OXYZ. Show that. OX'YZ is non-inertial. [6] | D2(522) 7 (Continued)

| : | (b) Prove that Lorentz transformation - : equations can be put in the form | , x") (cosh @ @ - sinh 6)/x | '(et' sinh @ cosh @)| + Hence show that Lorentz - transformations can be regarded as rotation through an imaginary | angle. . ". [6]

Question 3. (a) Prove that set of all - Lorentz transformations satisfies the group. ;

(a) properties. . [6] '(b) Determine the component velocities ' : . of a particle respect to two different frames of references S and S' using ; Lorentz transformation equations. [6] . 4 (a) 'By using Lorentz transformation equations, show that simultaneity |

(b) not absolute but relative. [6]

- (c) Calculate the length of a rod moving with a velocity $0.8c$ in a direction inclined at 60° to its own length. Proper length of the rod is given to be 100 cm .

Question 5. (a) Prove that three-dimensional

- (a) volume element $dx dy dz$ is not invariant. but four-dimensional, volume element $dx dy dz dt$ is ; ; invariant under Lorentz transformation. [6]
- (fb) If u and u' are velocities of a particle in two inertial systems S and S' respectively, where S' has velocity v , relative to S in x -direction, then - | show that— $|u'^2 + v^2 + 2u'vcos\theta|$ (Hose)

- (a) - : $(u'V \cos \theta)^2 : 1 + \frac{v^2}{c^2}$ where θ is the angle which w' makes with x -axis.. [6]

Question 6. (a) Find the velocity at which the mass of a particle is double its rest mass. [5]

- (a) Explain Michelson-Morley experiment, ; 7 7, Explain the following terms in detail : 12: fa) Light cone * ' (b) World points and World lines (ce) Space like intervals : (b) Time. like intervals. '